# Introduction to the history of computing



**Presented by**

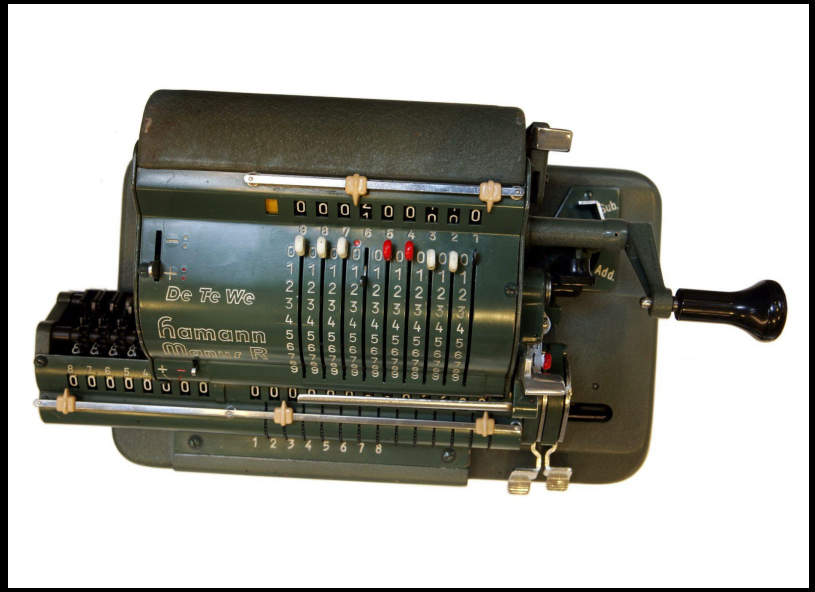
Cristian Camilo Cortes Ortiz

# Abacus

Computing originated to solve one of the main problems, which was calculation. The first computer/calculator was the abacus. The abacus was used to count (people, livestock, money, products, among other things) and keep track of accounts.



# Mechanical machines




Source

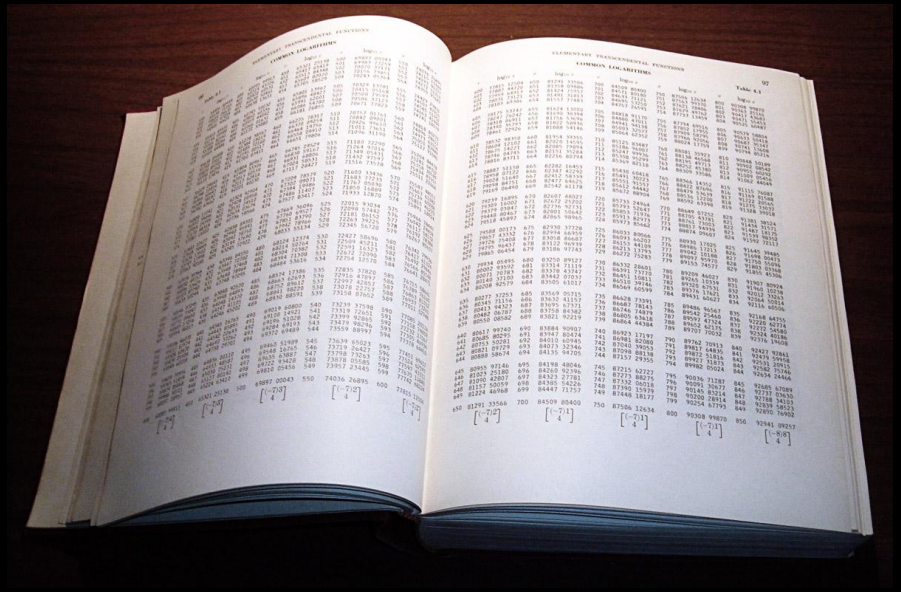
As the world evolved, calculations became more complex and more specific problems arose. The abacus evolved into non-electric mechanical machines that allowed certain resolutions to be reached with more complex calculations. These machines were a little more confusing to use than an abacus, but the calculations were more accurate.

# Human computers

This continued to evolve until it generated the term "computer," which was given to people who were dedicated to performing manual calculations and were known as "human computers." These people solved complex problems through manual calculation for various organizations (NASA, the military, large companies, etc.) that required precision.



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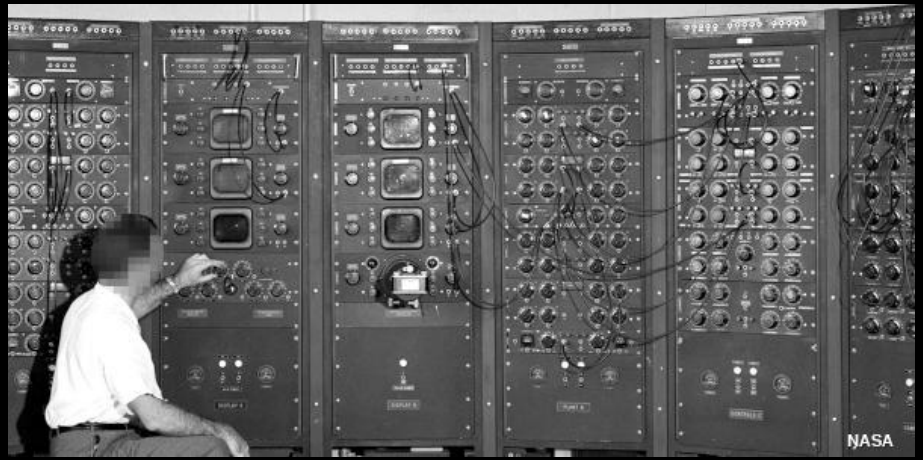
# Precomputation tables

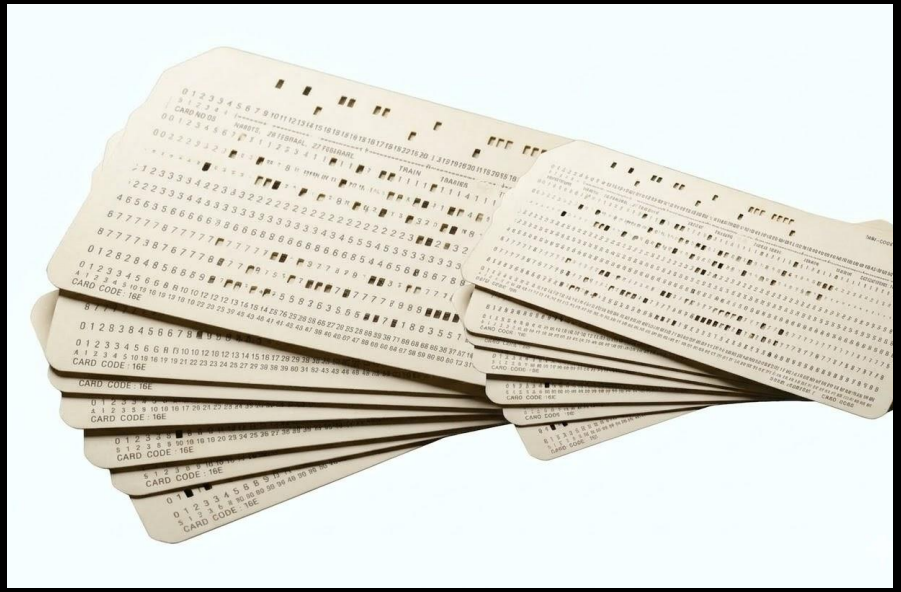
Solving these calculation problems was not scalable, nor was it particularly fast, as some problems required calculations that took longer than necessary to be solved by "human computers."

Therefore, aids for these "human computers" emerged in the form of books (precomputation tables) containing calculation tables, where one could look up and complete calculations to find a solution.

# Evolution

In the 1940s, there were increasingly specific problems that required greater precision in calculations, but above all, greater speed. Certain machines began to be developed that could be programmed little by little. Certain data was entered to generate the calculation, and the machine returned a result that helped with the calculation.



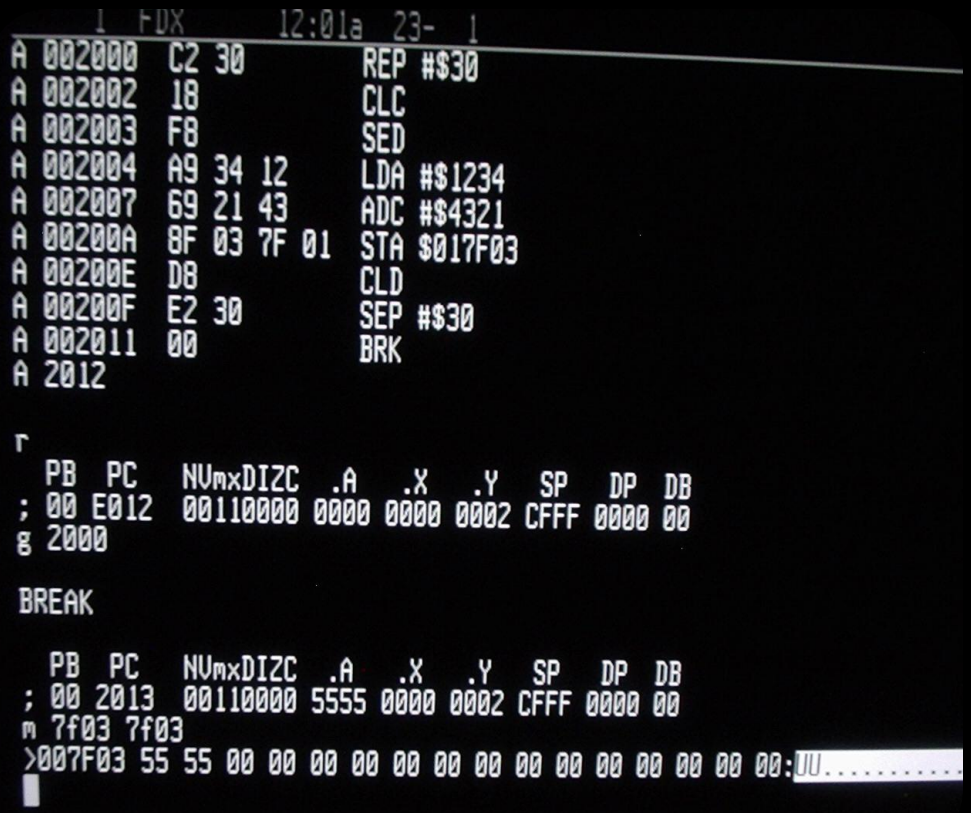


# Punch cards

As machines began to grow, a new problem arose: replicating calculations that had already been done previously. As a solution, punch cards were created. These cards were used to store information mainly for computer calculations.

# Machine code

In order to program machines without the need for punch cards and with the advancement of new technologies, machine code was developed. Machine code is the binary language used to program in 0s and 1s (zeros and ones) to give instructions to machines.

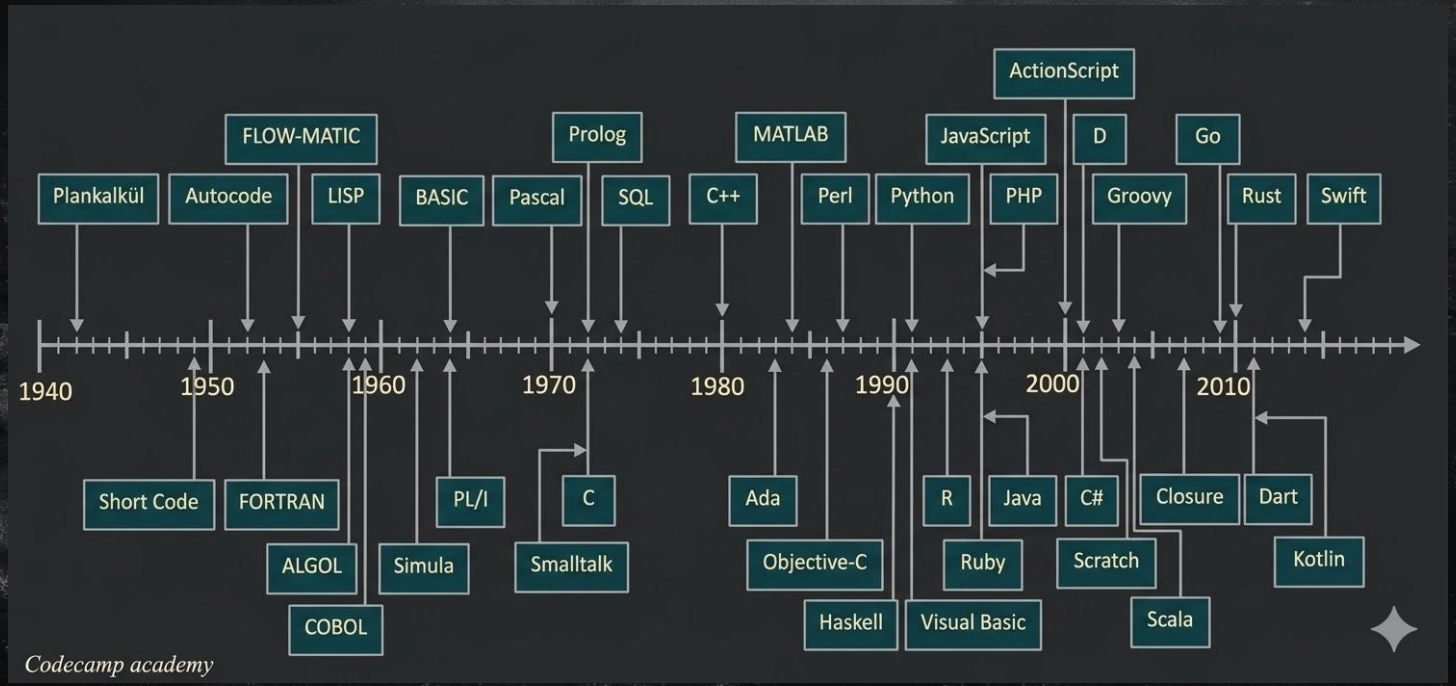


# History of computing

Programming in machine code was not very efficient, it was complex and took more time.

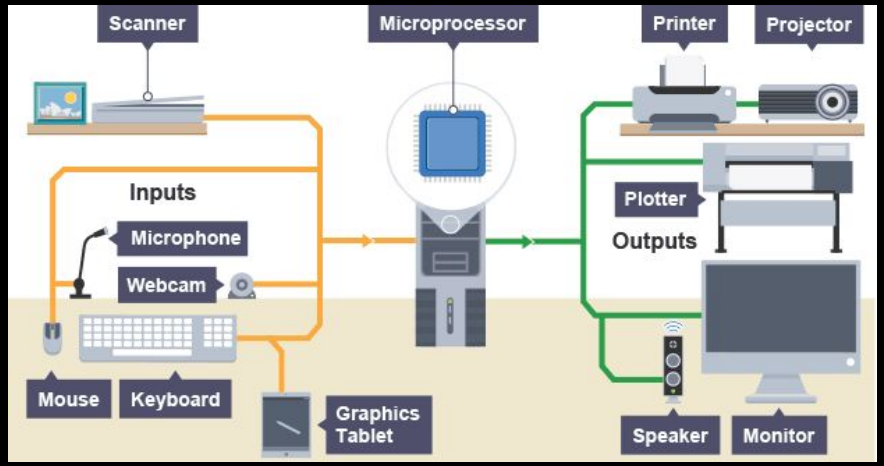
So the first programming languages emerged. These allowed programs to be written that people could read and understand more easily. These languages have a compiler, which is an intermediary that takes the code written by the programmer, compiles it, and converts it to machine code so that the machine can execute the instructions.

# History of computing



# Input and Output

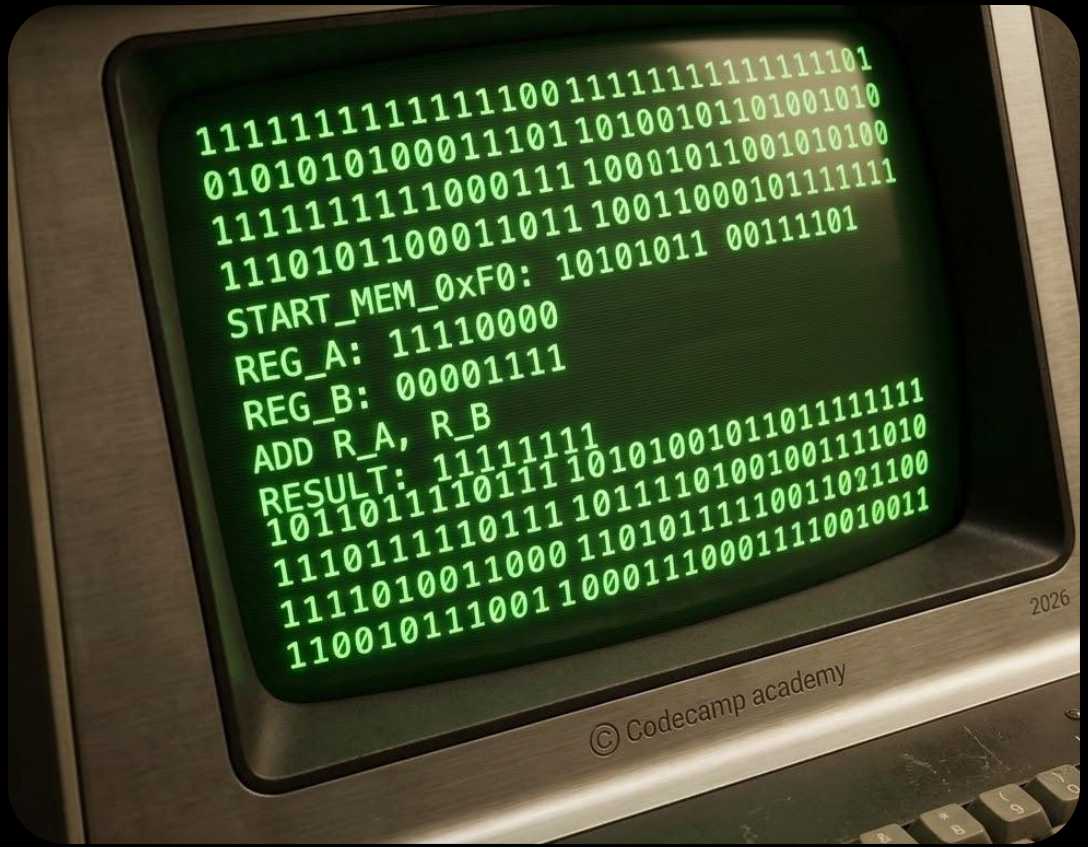
In the field of computer architecture, Input and Output (I/O) describes the essential communication bridge between an information processing system and the external environment. This cycle is the primary means by which a computer interacts with users, other hardware, or the physical world.



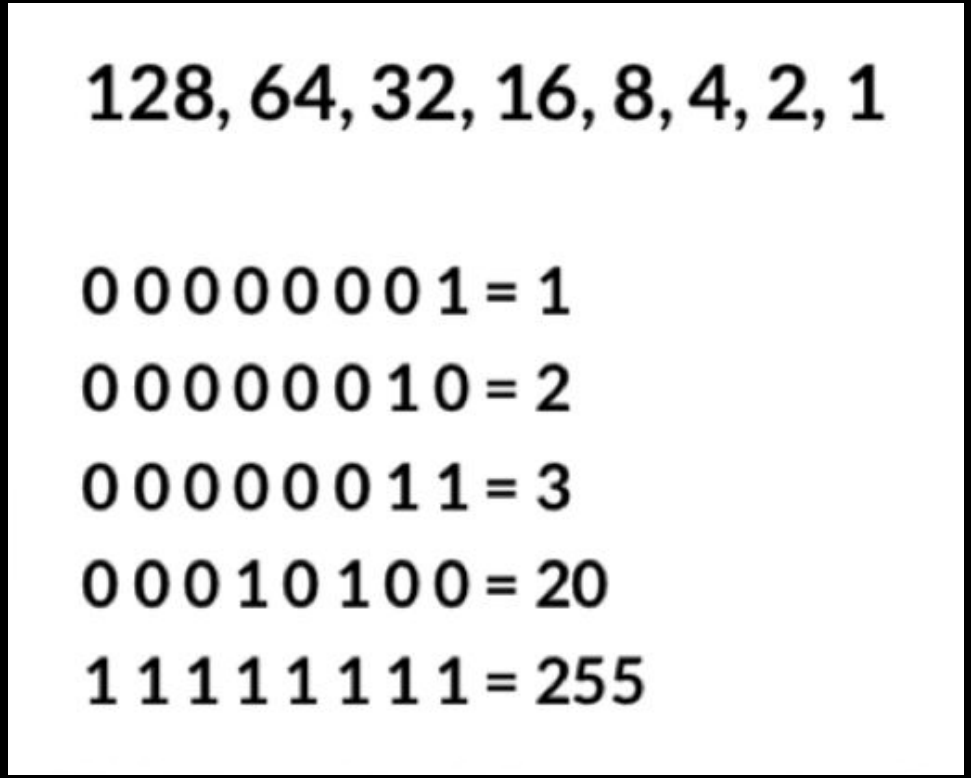

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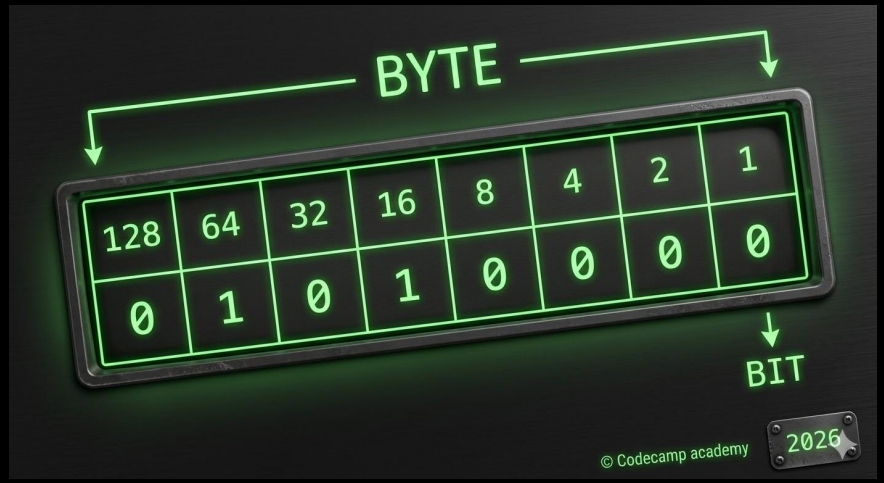
# Binary code

Binary code serves as the foundational language of all modern computing systems, utilizing a base-2 numerical system consisting solely of 0s and 1s. These digits, or "bits," represent the fundamental "off" and "on" electrical states within a processor's transistors. By organizing these bits into standardized sequences—such as 8-bit bytes—computers can encode complex information ranging from simple text characters to high-definition video. This elegant simplification allows physical hardware to execute abstract logic with incredible speed and precision.



# Binary code


128, 64, 32, 16, 8, 4, 2, 1
00000001 = 1
00000010 = 2
00000011 = 3
00010100 = 20
11111111 = 255




# BITs & BYTEs

The 0s and 1s ("zeros and ones") become bits and bytes. Each 0 and 1 ("zero and one") is a bit, and a combination of 8 0s and 1s ("eight zeros and ones") is a byte. This is how the computer receives the bytes and processes them directly to perform certain calculations to generate a response on an output device.

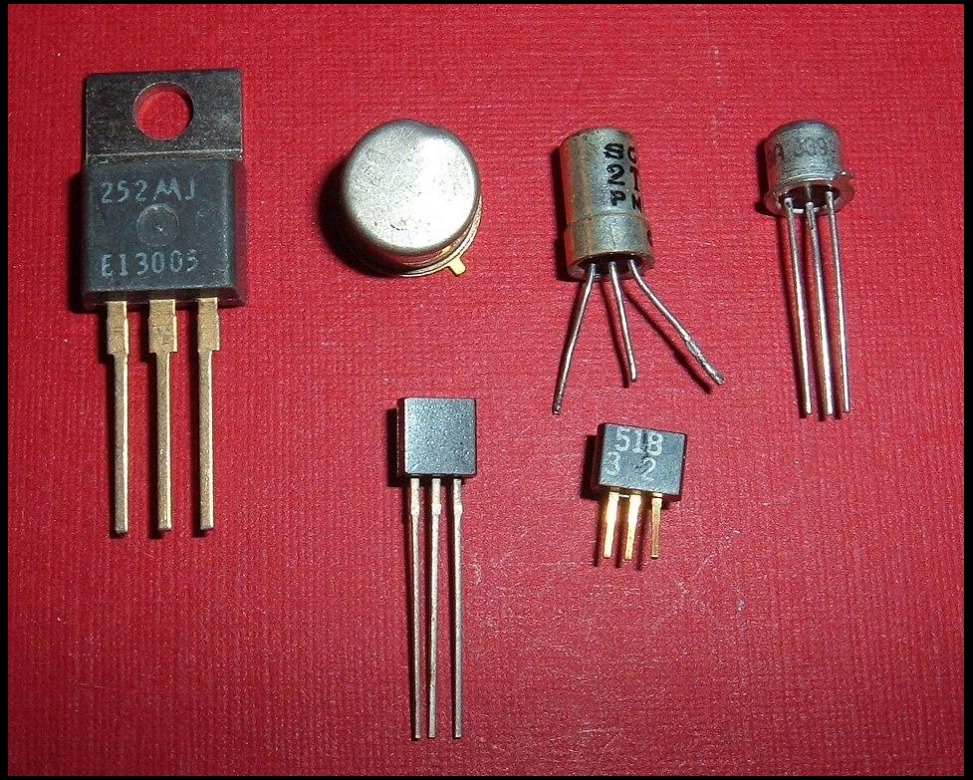
# Transistors

Transistors are one of humanity's greatest inventions and have contributed to the evolution of computers. Thanks to transistors, modern computing devices (laptops, desktops, smartphones, tablets, smart cars, robots, IoT, etc.) exist today.

Source



# Transistors




Source

# History of computing

Transistors operate in two states, OFF and ON, which can be represented as

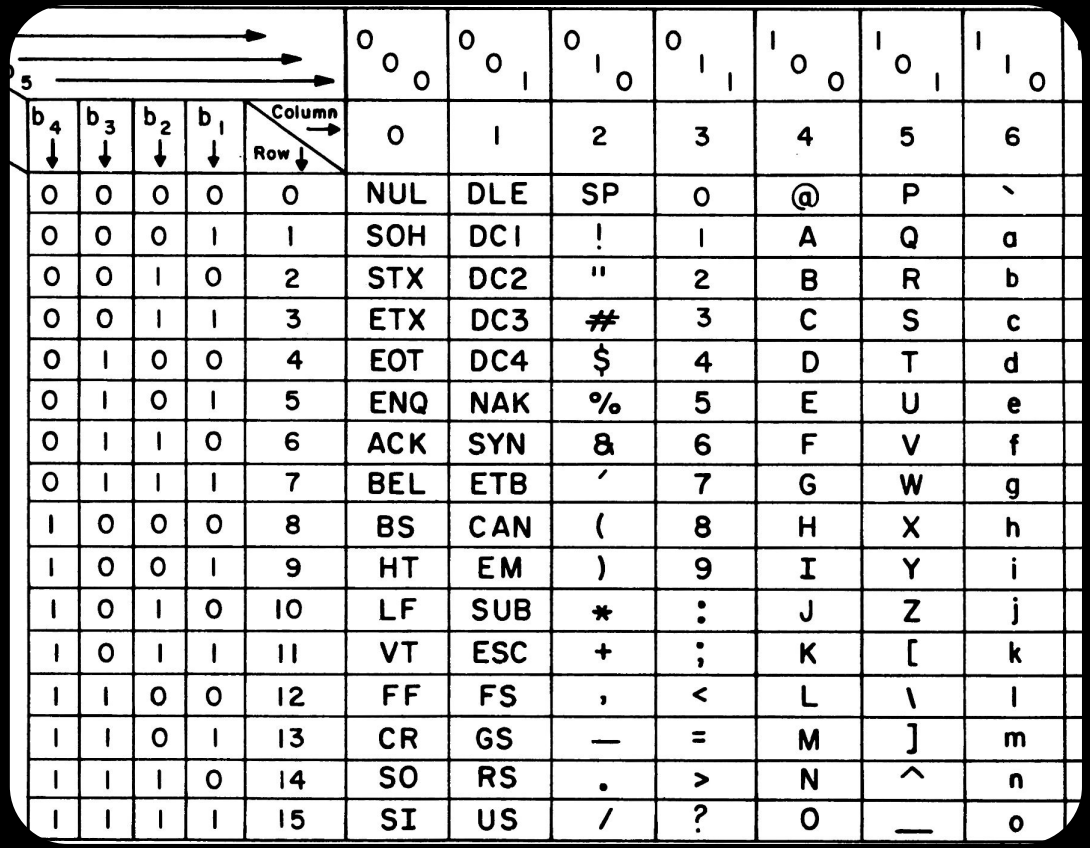



All bytes processed by the computer are processed as OFF and/or ON as a specific character, and the combination of bits and bytes are converted into logical instructions to be processed.

# ASCII

When the first programming languages emerged, the ASCII code (American Standard Code for Information Interchange) was created, which contains a table with characters, each of which corresponds to a number. When the computer processes a byte representing that number, it identifies what type of character it represents (letter, special character, number).

Source



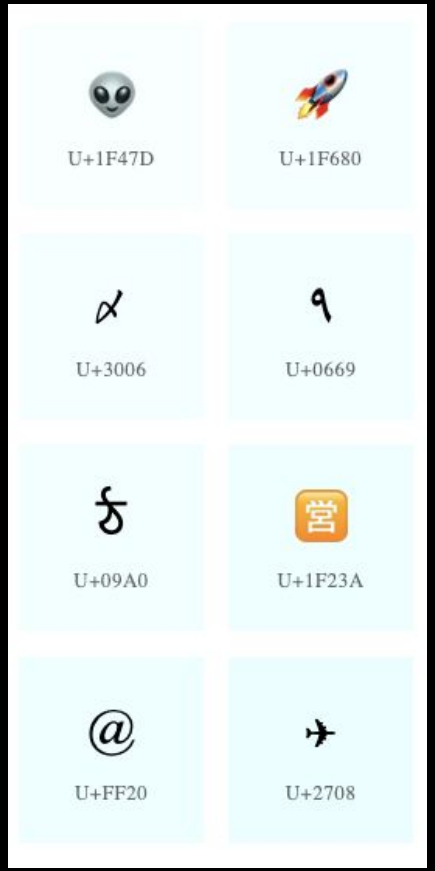
| b7 b6 b5 / b4 b3 b2 b1 ↓ | Column → 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| Row ↓ | 0 0 0 | 0 0 1 | 0 1 0 | 0 1 1 | 1 0 0 | 1 0 1 | 1 1 0 |
| 0 0 0 0 | 0 | NUL | DLE | SP | 0 | @ | P | ` |
| 0 0 0 1 | 1 | SOH | DC1 | ! | 1 | A | Q | a |
| 0 0 1 0 | 2 | STX | DC2 | " | 2 | B | R | b |
| 0 0 1 1 | 3 | ETX | DC3 | # | 3 | C | S | c |
| 0 1 0 0 | 4 | EOT | DC4 | $ | 4 | D | T | d |
| 0 1 0 1 | 5 | ENQ | NAK | % | 5 | E | U | e |
| 0 1 1 0 | 6 | ACK | SYN | & | 6 | F | V | f |
| 0 1 1 1 | 7 | BEL | ETB | ' | 7 | G | W | g |
| 1 0 0 0 | 8 | BS | CAN | ( | 8 | H | X | h |
| 1 0 0 1 | 9 | HT | EM | ) | 9 | I | Y | i |
| 1 0 1 0 | 10 | LF | SUB | * | : | J | Z | j |
| 1 0 1 1 | 11 | VT | ESC | + | ; | K | [ | k |
| 1 1 0 0 | 12 | FF | FS | , | < | L | \ | l |
| 1 1 0 1 | 13 | CR | GS | — | = | M | ] | m |
| 1 1 1 0 | 14 | SO | RS | . | > | N | ^ | n |
| 1 1 1 1 | 15 | SI | US | / | ? | O | _ | o |

# UNICODE

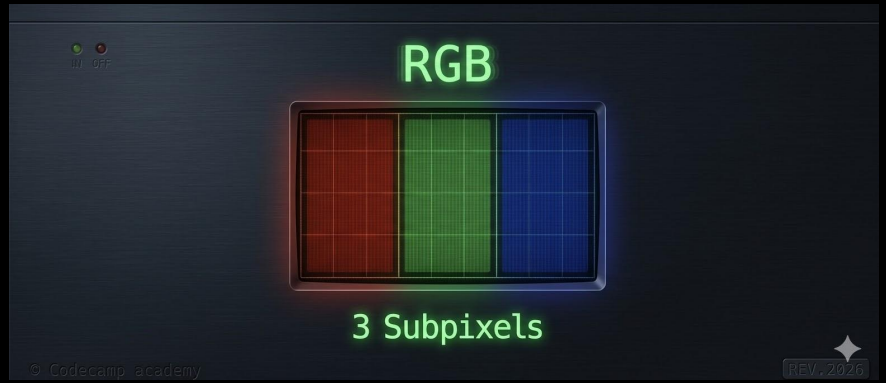
The ASCII code did not represent all existing alphabets and special characters. Therefore, the UNICODE code was created, which contains all existing alphabets and special characters.

An emoji is a combination of certain special characters that generate an image as a result. (Emojis are also part of the UNICODE code).

Each of the characters that make up the UNICODE code can be formed by 1 to 6 bytes.

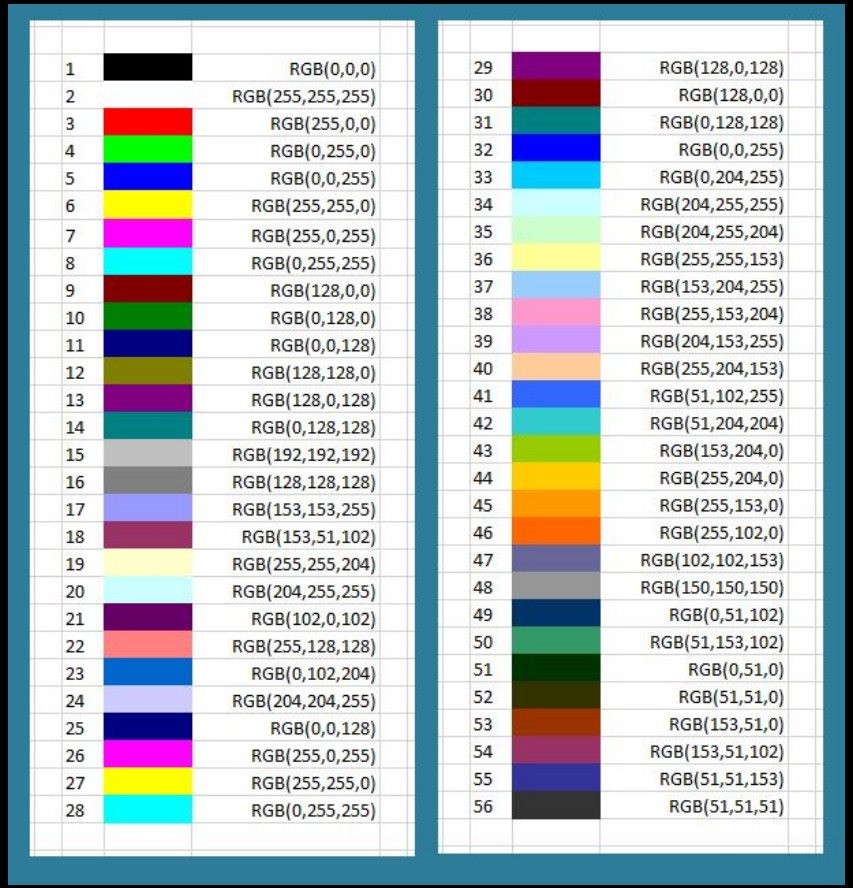



Source




# RGB

RGB (Red, Green, Blue) is a format that allows colors to be represented. Each of the pixels that make up an image is represented by 3 sub-pixel colors.



| | | |
|---|---|---|
| 1 | | RGB(0,0,0) |
| 2 | | RGB(255,255,255) |
| 3 | | RGB(255,0,0) |
| 4 | | RGB(0,255,0) |
| 5 | | RGB(0,0,255) |
| 6 | | RGB(255,255,0) |
| 7 | | RGB(255,0,255) |
| 8 | | RGB(0,255,255) |
| 9 | | RGB(128,0,0) |
| 10 | | RGB(0,128,0) |
| 11 | | RGB(0,0,128) |
| 12 | | RGB(128,128,0) |
| 13 | | RGB(128,0,128) |
| 14 | | RGB(0,128,128) |
| 15 | | RGB(192,192,192) |
| 16 | | RGB(128,128,128) |
| 17 | | RGB(153,153,255) |
| 18 | | RGB(153,51,102) |
| 19 | | RGB(255,255,204) |
| 20 | | RGB(204,255,255) |
| 21 | | RGB(102,0,102) |
| 22 | | RGB(255,128,128) |
| 23 | | RGB(0,102,204) |
| 24 | | RGB(204,204,255) |
| 25 | | RGB(0,0,128) |
| 26 | | RGB(255,0,255) |
| 27 | | RGB(255,255,0) |
| 28 | | RGB(0,255,255) |
| 29 | | RGB(128,0,128) |
| 30 | | RGB(128,0,0) |
| 31 | | RGB(0,128,128) |
| 32 | | RGB(0,0,255) |
| 33 | | RGB(0,204,255) |
| 34 | | RGB(204,255,255) |
| 35 | | RGB(204,255,204) |
| 36 | | RGB(255,255,153) |
| 37 | | RGB(153,204,255) |
| 38 | | RGB(255,153,204) |
| 39 | | RGB(204,153,255) |
| 40 | | RGB(255,204,153) |
| 41 | | RGB(51,102,255) |
| 42 | | RGB(51,204,204) |
| 43 | | RGB(153,204,0) |
| 44 | | RGB(255,204,0) |
| 45 | | RGB(255,153,0) |
| 46 | | RGB(255,102,0) |
| 47 | | RGB(102,102,153) |
| 48 | | RGB(150,150,150) |
| 49 | | RGB(0,51,102) |
| 50 | | RGB(51,153,102) |
| 51 | | RGB(0,51,0) |
| 52 | | RGB(51,51,0) |
| 53 | | RGB(153,51,0) |
| 54 | | RGB(153,51,102) |
| 55 | | RGB(51,51,153) |
| 56 | | RGB(51,51,51) |



# RGB

Each of these colors is in the range of 0 to 255. Depending on the hue of each of these three colors that make up a pixel, that is the color that the pixel will represent.

Source

# RGB colors

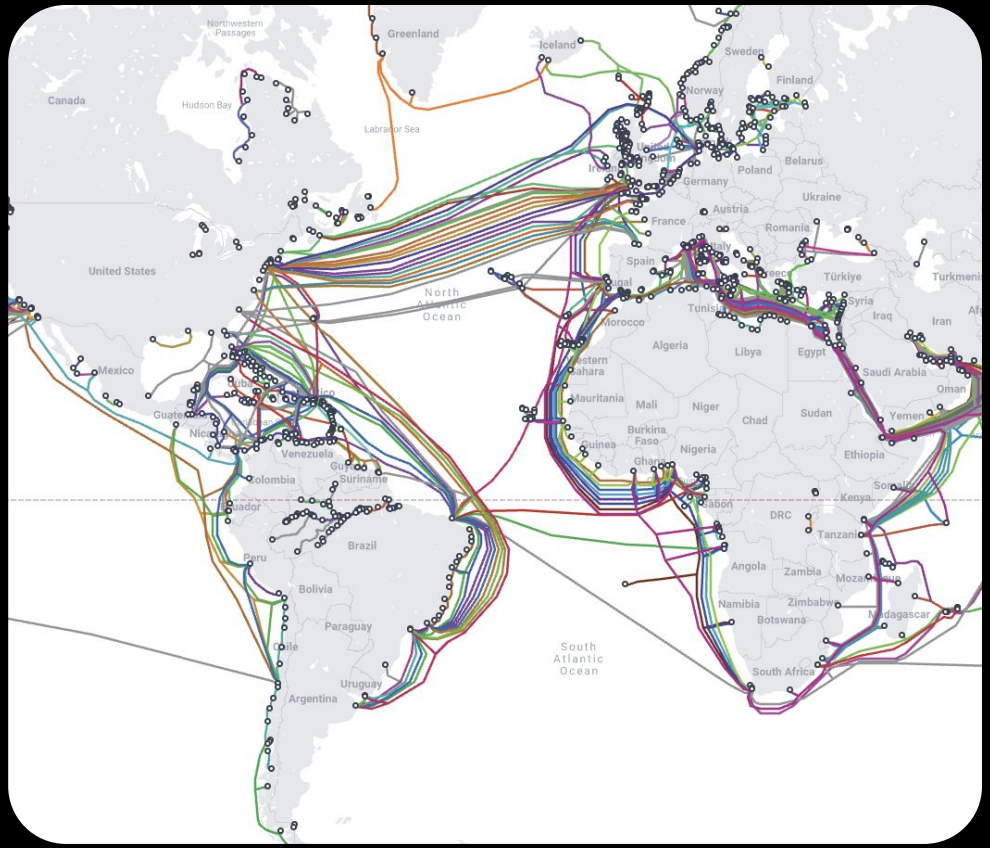
R: 178
G: 80
B: 228

R: 178 = 10110010
G: 80 = 01010000
B: 228 = 11100100



# Internet

With the evolution of operating systems and programming languages, there is a need to connect computers and exchange information. This has led to the emergence of the internet, which consists of fiber optic cable connections that connect the entire world.




Source

# IoTs

The Internet of Things (IoT) represents a sophisticated paradigm in computing where physical devices such as the sensors, thermostat, smart lock, and industrial vibration monitor visualized here are seamlessly integrated into a digital network. This ecosystem relies on a central IoT Gateway or Edge Computing Hub to act as the intelligence core, as depicted by the central module with its screen showing the device icons. These components communicate via an intricate network of data pathways (the green circuit traces), transmitting real-time data to create a synchronized Device Interconnectivity and Sensor Network.



# Thank you!

